

Lauren Kimpel

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EDUCATION

2024 - present PhD in Mathematics, **The University of Delaware**
2021 - 2024 M.S. in Applied & Computational Mathematics, **Johns Hopkins University**
2016 - 2020 B.S. in Mathematics & Computer Science, **Temple University**

RESEARCH

Banaian, Esther et al. (2024). “The e-positivity of the chromatic symmetric function for twinned paths and cycles”. In: *Discrete Mathematics*. URL: <https://api.semanticscholar.org/CorpusID:270067954>.

Kimpel, Lauren (2024). “Some Fellow-Traveler Properties on Finite Graphs”. MA thesis. Johns Hopkins University.

Cyber Applied Research Scientist

Jan 2021 - Aug 2024

Lockheed Martin Advanced Concepts Laboratory

Cherry Hill, NJ

- Played a key role in developing a novel network optimization and reasoning framework, resulting in an accepted research proposal to a U.S. DoD customer, and securing funding for a multi-year program
- Performed static and dynamic reverse engineering on x86/x86_64, SPARC, and PowerPC architectures. Mapped memory & control flow. Developed prototypes for cyber capabilities.
- Responsible for literature reviews and whitepaper writeups.
- Prototyped solutions related to integer programming, limited-information reasoning and symbolic execution. Wrote Python/C simulations.

Undergraduate Research Assistant

May - Sep 2020

Computer & Information Sciences Department, Temple University

Philadelphia, PA

- Analyzed the Pufferfish differential privacy framework for concrete tradeoffs between privacy levels and data freshness when sanitizing sensitive time series data
- Developed 2-state Markov chain-based mechanisms to probabilistically account for vulnerabilities related to data freshness metrics in ϵ -Pufferfish Private algorithms

TEACHING

Graduate Instructor of Record

January 2026

Department of Mathematical Sciences, University of Delaware

Newark, DE

- MATH 010 (Intermediate Algebra); Winter 2026

Graduate Teaching Assistant

Aug 2024 - May 2026

Department of Mathematical Sciences, University of Delaware

Newark, DE

- MATH 231 (Integrated Calculus 1A); Fall 2025
- MATH 115 (Precalculus); Fall 2025
- MATH 205 (Statistical Methods); Spring 2025, Spring 2026
- MATH 241 (Analytic Geometry and Calculus A); Fall 2024

Undergraduate Teaching Assistant

Aug - Dec 2020

Computer & Information Sciences Department, Temple University

Philadelphia, PA

- CIS 3223 (Data Structures & Algorithms); Fall 2020

CONFERENCES AND WORKSHOPS

Graphical Models in Algebraic Combinatorics Summer School

Simons Laufer Mathematical Sciences Institute

Graduate Research Workshop in Combinatorics

The University of Wyoming

June - July 2025

Berkeley, CA

July 2023

Laramie, WY

PROJECTS

MATH 637 Final Project: Adversarial Examples for Image Classification Networks 2025

- Tested FGSM (Fast Gradient Sign Method) attacks against two victim image classification neural networks using black box techniques (training a surrogate CNN to learn from the output of the victim) and white box techniques (direct FGSM.)
- Uses gradient-based perturbation methods to “force” an image classifier to misclassify images at test-time

Puffin: Simplicial Path Complexes for Obfuscation Detection 2023

- Transformed the control flow graphs (CFGs) of 12 binaries into 3-simplicial path complexes, using a filtration distance based on the combinatorial distance metric (maximum-length shortest path) between two vertices.
- Binaries were obfuscated by inserting extra bytes into the assembly code (.s files.) These are interpreted by the assembler to be extra JMP instructions, which causes the CFG to admit a more complex structure.
- Analyzed the homological differences between the unobfuscated and obfuscated binaries, and used the resulting data to train a Random-Forest Classifier to identify, given an arbitrary binary, whether it had been obfuscated using the JMP-insertion technique.

MENTORSHIP

Graduate Mentor

University of Delaware Directed Reading Program

September-December 2025

Newark, DE

- Responsible for regularly giving suggested readings and research guidance to a second-year undergraduate student in graph theory, and, where applicable, coming up with interesting and challenging problems appropriate to the background(s) of both mentor and mentee, culminating in a final presentation.
- Fall of 2025: studied planar graphs, chromatic polynomials, coloring, and Kuratowski’s Theorem (the forbidden subgraph problem).

VOLUNTEERING

Member, Association for Women in Mathematics (Student Chapter) 2025

Presenter, Young Women’s Conference in Science, Engineering, & Mathematics 2023

Princeton University

OTHER EXPERIENCE

Multidisciplined Linguist Intern

The United States Department of Defense

May - Aug 2019

Fort Meade, MD

- Obtained Top Secret security clearance and passed lifestyle polygraph
- Analyzed Russian morphologic, syntactic and phonetic structures in graphical and voice material to determine subtext, meaning and intention for high priority analysis and production mission critical to IC community customers, including senior U.S. decisionmakers

SKILLS

Programming Languages	Python, C, R, MATLAB, L ^A T _E X, Java
Assembly	x_86/x_86-64, PowerPC
Reverse Engineering & Debugging Tools	Ghidra, WinDbg, gdb, Wireshark
Data Visualization & Analysis	networkX, matplotlib
Foreign Languages	English (native), Russian ¹
Other	Atlassian Tools, Git, Windows, Linux, Emacs, SageMath

Last updated: December 26, 2025

¹can read on a basic level and write on a slightly more intermediate level; can speak but with nontrivial levels of impairment.